

EXERCISES

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1. Introduction to R

1. EXPRESSIONS

An expression in R corresponds to a sequence of operations, and it can include variables and/or functions. These expressions can be evaluated in R and have (only one) return.

Exercise 1.1

Consider the following expressions:

- $123 * 321$
- $42 ** 2$
- $34 - 2 * 6$
- $4 + 2 == 42$
- $1 + \text{sum}(2, 3)$

- a) Assess the result of these expressions in R.
- b) Save the result of the last expression in a variable named x .
- c) Use that same variable, x , in another expression.
- d) Attribute to the variable x the value 42. What happened to the result of the expression in b)?

Exercise 1.2

Assess the result of the following expressions in R:

- $(-2)^4$
- $12 + 5 \times \pi$
- $3^{7-4} + \sqrt{2}$
- $\sqrt{(5-2)^2 + (7-3)^2}$
- $\frac{20 + 18 + 24}{3}$
- $\sqrt[3]{(20 \times 8 \times 24)}$

Exercise 1.3

Consider the Table below. Save the expressions of the column *Expression* in the corresponding variables, indicated in the column *Variable*.

Table 1: Variables

Variable	Expression
a	1
b	-5
c	6
A	$\pi \times 6371^2$

Exercise 1.4

Use the variables defined in the previous question to assess the following expressions:

- $A == a$
- $A = a$
- $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

2. VECTORS

Vectors are a simple yet powerful way to store information in R. However, these objects can only store one type of information (i.e., numeric, character, logical).

Exercise 2.1

Consider the following Table:

Table 2: Sales Performance

Salespeople	Sales
Alice	\$8,500
Bob	\$7,200
Charlie	-
David	\$9,000
Francis	\$6,300

-
- a) Save the information in the Table 2 in two separate vectors: *salespeople* and *sales*. Beware of the missing grade. What would happen if you saved the data in the same vector.
 - b) Access the name and sales of the 5th salesperson.
 - c) Charlie's sales were updated, he had \$8,000. Update his sales on the vector *sales*.
 - d) Calculate the mean of the sales.
 - e) The manager decided to attribute a bonus to all salespeople. Add \$200 to each salesperson's sales.
 - f) Access the sales of the first three salespeople.
 - g) List the names of all salespeople except of the last 3.
 - h) List the salespeople who achieved sales greater than or equal to \$8,200.
 - i) List the salespeople who achieved sales between \$7,000 and \$8,800.
 - j) Return the sales of 4 salespeople chosen randomly (*Hint*: use the function *sample()*).
 - k) Combine the *salespeople* and *sales* vectors into a single named vector called *salespeople_sales*. Set the names of the vector elements to be the salesperson names. This way, each salesperson's name is associated with their respective sales.

Exercise 2.2

Consider the following vectors:

$a = c("L", 3, 3, T)$ and $b = c(F, 4, 1, 7)$

Explain what happened to the values saved in both vectors.

3. MATRICES

A matrix, in R, is an extension of a vector to two dimensions.

Exercise 3.1

Consider again the Table 2.

- a) Create a new vector, *performance*, indicating the salesperson's performance ratings (*Note*: You can assign the performance ratings randomly).
- b) Construct a matrix named *sales_data* with the sales from Table 2 and the vector *performance*.
- c) Use the vector *salespeople* to label the lines of the matrix *sales_data*.
- d) Swap the lines by the columns in the matrix *sales_data*.

Exercise 3.2

Consider the following vector that contains sales data:

```
sales2 = sample(5000:10000, 20)
```

Transform it into a matrix *sales_data2*, with 5 lines and 4 columns.

Exercise 3.3

Add labels to the columns and lines of the matrix *sales_data2*.

Exercise 3.4

Access the elements of the matrix *sales_data2* in different ways:

- By two positions.
- By two names.
- By one position and one name.
- A whole line by a position.
- A whole column by a name.
- The first three lines and all the columns except the last one.

Exercise 3.5

Calculate the mean of values for each line and each column.

Exercise 3.6

Assess the result of some operations on the following matrix:

```
ma = matrix(as.integer(runif(16, 1, 20)), 4, 4)
```

- `ma + 1`
- `ma * 2`
- `ma * ma`
- `ma %% ma`

4. DATA FRAMES

The structure of `data.frame` is one of the most used in R, mainly due to its versatility in the type of objects supported and in indexation: it is derived from lists, supporting elements from different types in a single object, and it adds the functionalities of indexation of vectors.

It is mainly used to keep tables of data, being similar to a matrix with names in lines and columns, but it supports elements from different types. This type of table can be seen as a database, or a spreadsheet, where each line corresponds to a different register.

Exercise 4.1

Transform the matrix `sales_data2`, created in the Matrices section, into a `data.frame`.

Exercise 4.2

Check the dimensions of the data frame created.

Exercise 4.3

Show the lines where salespeople had all sales higher than \$6,000.

5. FACTORS

Exercise 5.1

We have collected data on the regions where our sales representatives operate. Create a factor called *region* from:

```
c("north", "south", "south", "south", "east", "west", "south")
```

a) Verify the levels created.

b) With another group of sales representatives, the following vector was registered:

```
c("south", "south", "west")
```

Define a factor for this vector but keep the same levels as in a).

c) Considering again the levels from a), transform into a factor the data from a third group of sales representatives:

```
c("west", "south", "central", "south")
```

What happened to the third element?

-
- d) In the vector *gen*, we have registered the gender of the sales representatives:

```
gen = c("m", "f", "m", "m", "f", "f", "m")
```

Create a contingency table, showing the gender by region.

- e) Select the first three elements from the factor *region*. What happened to the levels? How can we keep only the levels from the selected elements?

- f) Finally, we have registered the performance of the sales representatives in the vector *performance*. This information is qualitative, but it has order.

```
performance = c("poor", "poor", "good", "excellent", "excellent", "good", "good")
```

Transform this vector into a factor while maintaining the order relationship of the elements.

Compare the performance between two sales representatives.

6. LISTS

Lists, opposed to vectors and matrices, allow to keep elements of different types (i.e., numeric, character, logical) in a single object. Hence, lists can be viewed as an ordered collection of objects. The objects that constitute a list are called components, they are numbered and can have a label associated with them.

Exercise 6.1

Create the list *employee* to represent a record containing the following details:

id 1999001

name John Smith

sales \$8,700; \$7,300; \$9,500; \$7,000; \$8,200

employed yes

Exercise 6.2

Visualize all the components (or attributes) from the list. Compare with the results from the function *str()*.

Exercise 6.3

Access the first component of the list, using the same syntax as in vectors. Verify the type of object of that result, using the function *typeof()*.

Exercise 6.4

You have received additional information about the employee: his monthly salary is \$50,000. Update the record with that information.

Exercise 6.5

Add the following information only through one step:

gender male

managers Emily White and Michael Davis

Exercise 6.6

The employee had another sale of \$10,000. Add that sale to the employee's record.

Exercise 6.7

Delete the *salary* component from the *employee* list.

Exercise 6.8

How many components does the *employee* record have now?